

Weather-Aware Predictive Optimization of LEO Satellite Passes Using Calibrated Random Forests

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Abstract—This study addresses the impact of meteorological phenomena as a significant source of uncertainty and vulnerability in LEO satellite communications. A predictive model of satellite passes is developed that incorporates the influence of variable weather conditions, using supervised learning techniques based on a calibrated Random Forest classifier to estimate the probability of successful communication during each satellite pass. The proposed model is then integrated into an optimization scheme with the aim of maximizing transmission speed, improving the efficiency of the link between the satellite and the ground station. The performance of the proposed optimization model is evaluated in terms of transmission speed and quality of service, comparing it with conventional methods. The results show that the proposed method outperforms heuristic methods by using calibrated probability estimates to optimize planning, resulting in more robust and efficient ground station operations.

Keywords—Internet of Things (IoT); low Earth orbit (LEO); satellite, optimization

I. INTRODUCTION

The space sector and sixth generation of wireless communication systems (6G) are undergoing unprecedented transformation, driven by the development of satellite megaconstellations by private initiatives such as Starlink and OneWeb [1], [2]. Specifically, many of the services developed by these private initiatives have focused on low Earth orbit (LEO) satellites, which offer fundamental advantages such as lower latency and higher data transmission speeds. These characteristics make them ideal for both applications that require real-time communication and devices with limited capabilities, such as the IoT [3], [4]. Unlike geosynchronous Earth orbit (GEO) satellites, which remain fixed over a specific region, LEO satellites operate at much lower altitudes, typically between 180 and 2000 km above the Earth's surface, providing more dynamic and flexible coverage globally [5].

However, LEO satellites require ground stations (GS) networks and the satellites themselves to function as a dynamic, integrated system to ensure continuous connectivity [6]. Furthermore, to achieve truly global coverage in regions with little or no ground infrastructure, these constellations are essential [7]. Therefore, planning and scheduling satellite passes at GS is a critical challenge that must be resolved to improve the reliability of space communications.

Adding to this complexity is meteorological uncertainty, which is a critical factor in satellite communications [8]. While LEO constellations offer some resilience to physical ground infrastructure failures, thanks to their proximity to GS and

high transmission rates, communications remain vulnerable to atmospheric phenomena. Signal attenuation due to rain is one of the main challenges, especially in systems operating in high-frequency bands. Most commercial LEO constellations, such as OneWeb, use the Ka (26.5–40 GHz) and Ku (12–18 GHz) bands, which are highly sensitive to absorption and scattering by raindrops [9]. In the case of Starlink, in addition to these bands, intersatellite laser optical links are used, which increase capacity and reduce latency, although microwave links remain vulnerable to rain attenuation [10], [11].

The severity of this attenuation depends directly on the raindrops and the intensity of the precipitation. In tropical and equatorial regions, where heavy rains are frequent, this phenomenon can significantly degrade signal quality, causing data loss or even complete failure of scheduled communications [12], [13]. Traditional scheduling approaches, based on fixed constraints and deterministic visibility windows, do not adequately incorporate the stochastic nature of meteorological uncertainty. This leads to inefficient use of system resources, as a scheduled window may be unusable in practice, turning what could have been a successful pass into a failed one. Therefore, it is necessary to address an instantaneous predictive model for satellite pass scheduling under meteorological uncertainty with robust optimization techniques for LEO communications scheduling. Addressing this challenge is essential for improving the efficiency of next-generation space communications networks.

A. Related Work

The prediction of trajectories for satellites in LEO by an GS and the scheduling of tasks constitute a resource allocation problem that has been extensively researched by various authors [14]–[19]. Task allocation has been shown to be a complex problem due to the hardware and software limitations of LEO satellites, as they are generally CubeSats. Among the most critical tasks are downloading data to the GS, observation and data acquisition, executing telemetry commands for adjustments, and monitoring the status of the satellites [10], [20], [21].

To address problems related to satellite trajectories, closed-loop machine learning-based approaches have been proposed to predict the orbit of LEO satellites, with the aim of improving navigation [22]. These systems combine learning models with real-time feedback, achieving a more accurate prediction of satellite positions relative to an GS, which is essential for opportunistic navigation. However, according to studies, uncertainties arising from changing weather conditions are not

considered, nor is this information used as a strategy for task scheduling.

Atmospheric phenomena such as rain, clouds, and ionospheric variations directly affect satellite communication channels and, consequently, the quality of radio signals transmitted by satellites or GS [23]. In addition, techniques such as machine learning (ML) and deep learning (DL) are known to offer a promising alternative for modeling these atmospheric phenomena in real time [24]. Recent studies show that the use of models based on artificial neural networks (ANN) and long-term memory (LSTM) can predict attenuation and precipitation rates with great accuracy [25]. Similarly, the use of 3D convolutional neural networks (3D CNN) has been proposed to predict rain attenuation events using spatiotemporal data from satellite and radar images [26].

This advanced prediction capability is essential for systems that require reaction time to execute proactive actions, such as task planning aimed at maintaining service quality. Furthermore, [27] proposes the use of DL to predict channel status in adverse weather conditions in satellite communication systems LEO that employ multiple-input multiple-output (MIMO). The result is a channel predictor that effectively mitigates the problem of channel aging in such systems. Therefore, improving the accuracy of satellite trajectories under meteorological uncertainty can significantly contribute to optimizing task scheduling by formulating an optimization problem aimed at maximizing throughput metrics.

On the other hand, studies on optimization methods in LEO satellite networks conclude that the main aspects to consider are pass scheduling, coverage, costs, and quality of service [28]. In this context, recent research has incorporated on-orbit edge computing (OEC) in LEO satellites to address these task scheduling challenges. This new approach reduces energy consumption by up to 40% in GS compared to reference solutions [29].

Furthermore, OEC facilitates the application of optimization approaches based on Mixed Integer Linear Programming (MILP), as well as methods such as genetic algorithms (GA), simulated annealing (SA), and other metaheuristics widely used in the field of LEO satellites [28], [30]. A NASA study on the scheduling of Earth observation satellite constellations concluded that simulated annealing was the most effective search algorithm, and that integrated constellation scheduling was superior to individual satellite scheduling [31], [32].

B. Novelty and Contributions

Given the good results obtained in previous studies [10], [14]–[19], [26], [32], this study addresses the impact of meteorological phenomena as a significant source of uncertainty and vulnerability in satellite communications LEO. To this end, a predictive model of satellite passes is developed that incorporates the influence of variable meteorological conditions. The proposed model is subsequently integrated into an optimization scheme aimed at maximizing the transmission rate, improving the efficiency of the link between the satellite and the GS. The main objective of this work is to design a system capable of integrating satellite pass prediction under

meteorological uncertainty into the resource scheduling and optimization process, in order to maximize throughput metrics and improve the quality of service perceived by the end user. To achieve this purpose, the work is structured around the following specific objectives:

- To model Two-Line Elements (TLE) data of LEO satellites through the public Celestrak API, which allows access to historical data and the calculation of key satellite parameters at any given time.
- To model meteorological data through the public Open-Meteo API, identifying atmospheric variables such as precipitation and cloud cover that directly influence the quality of satellite communication links.
- To develop a predictive model of satellite passes that integrates orbital data TLE with meteorological conditions, using supervised learning techniques based on a calibrated Random Forest classifier to estimate the probability of successful communication during each pass.
- To integrate the predictive model into an optimization framework based on MILP, aimed at efficient task scheduling and optimal resource allocation, with the objective of maximizing the effective transmission rate.
- To analyze the metrics of the predictive model and evaluate the performance of the proposed optimization model in terms of transmission rate and quality of service, comparing it with conventional methods.

II. SYSTEM MODEL

The system model consists of three elements, as shown in Fig. 1. The first element is the LEO satellites, which forms part of a constellation and acts as an orbiting node, transmitting and receiving information during its windows of visibility over the Earth. The second element, the communication channel, connects the satellite to the GS and is subject to both the geometric conditions of the orbit and the influence of the atmosphere, where meteorological factors introduce uncertainty into the quality of the link. Finally, the GS receives the data sent from the satellite and transmits it to the ground infrastructure, where planning and optimization strategies are applied. This simplified model serves as the basis for the study.

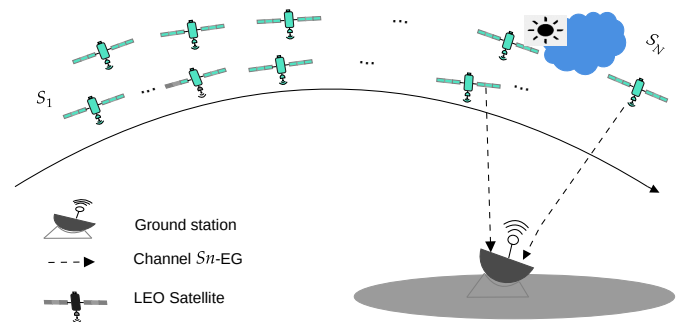


Fig. 1. System Model for Predictive Scheduling of Satellite Passes in LEO Constellations under Meteorological Uncertainty

Within the visibility zone, calculations can be performed to determine certain parameters. For example, the slant range is

determined as follows:

$$d = \|\mathbf{r}_s - \mathbf{r}_g\|, \quad (1)$$

where $\mathbf{r}_s$ is the position vector of the satellite in Earth centered Earth fixed (ECEF) coordinates and $\mathbf{r}_g$ is the position vector of the GS in the same system. To calculate the observation angles, the rotation of the ECEF system is applied to the local east north up (ENU) system:

$$\begin{pmatrix} e \\ n \\ u \end{pmatrix} = \mathbf{R}_{\text{ECEF} \rightarrow \text{ENU}}(\phi_g, \lambda_g) (\mathbf{r}_s - \mathbf{r}_g), \quad (2)$$

and

$$\mathbf{R}_{\text{ECEF} \rightarrow \text{ENU}}(\phi_g, \lambda_g) = \begin{bmatrix} -\sin \lambda_g & \cos \lambda_g & 0 \\ -\sin \phi_g \cos \lambda_g & -\sin \phi_g \sin \lambda_g & \cos \phi_g \\ \cos \phi_g \cos \lambda_g & \cos \phi_g \sin \lambda_g & \sin \phi_g \end{bmatrix} \quad (3)$$

where ϕ_g and λ_g are the latitude and longitude of the GS. The link budget is determined as follows:

$$P_r(\text{dBm}) = P_t(\text{dBm}) + G_t(\text{dBi}) + G_r(\text{dBi}) - \text{FSPL}(\text{dB}) - L_{\text{atm}}(\text{dB}) - L_{\text{misc}}(\text{dB}), \quad (4)$$

where $P_t(\text{dBm})$ is the transmit power, $G_t(\text{dBi})$ is the transmitter gain, $G_r(\text{dBi})$ is the receiver gain, FSPL(dB) is the free-space path loss, L_{atm} represents atmospheric attenuation and L_{misc} includes other losses. To simplify the modeling process, data and parameters were obtained using the Celestrak API and Open-Meteo API, as the limitations of the study prevented the deployment of a real experimental environment.

III. METHODOLOGY

The methodology developed in this work is based on four main steps (see Figure 2): the construction of the dataset of satellite passes over a GS, the development of the predictive model of the probability of success of each pass over the GS, the formulation of the optimization problem, and the comparison with the reference policies. First, TLE data were collected from the Celestrak API using the Skyfield library [33]. In addition, time series of precipitation and cloud cover

orbital characteristics such as the duration of the pass and the maximum elevation of the satellite were extracted, and from the meteorological data, the average and maximum precipitation and the average and maximum cloud cover were calculated.

Given that this study was conducted in a simulation environment, an indirect labeling process known as proxy labeling was adopted. This procedure consists of defining rules based on physical knowledge of the system to classify each pass as successful or unsuccessful, using criteria that approximate expected performance in real scenarios. Two main variables were considered for the construction of the labels: the maximum elevation reached by the satellite during the pass and the average precipitation forecast for the corresponding interval.

According to [35], in Ku and Ka band links, the elevation angles required to ensure acceptable signal levels at the receiver are generally between 10° and 30° . Additionally, operational networks typically use 20° as the minimum practical threshold to ensure stable downlinks in commercial and observational applications [36]. For this reason, in this work, passes with a maximum elevation of less than 20° were considered potentially unfeasible.

On the other hand, in modern systems operating between 2 and 40 GHz, precipitation values greater than approximately 0.5 mm/h are associated with rain fade events, capable of significantly degrading the received signal, especially in tropical regions [37]. Consequently, the criterion was adopted that passes with average precipitation greater than 0.5 mm/h have a high probability of failure. Based on these criteria, each pass was labeled as follows:

$$\text{label} = \begin{cases} 1, & \text{if maximum elevation} > 20^\circ \\ & \text{and average precipitation} < 0.5 \text{ mm/h} \\ 0, & \text{otherwise} \end{cases} \quad (5)$$

Furthermore, to reflect model uncertainty and real channel oscillations, a random noise component of 5% was added.

A Random Forest classifier with 200 trees was created on this set of data, calibrated using CalibratedClassifierCV with stratified cross-validation of five partitions. This calibration allowed us to obtain more realistic estimates of the probability of success for each pass, which will later be used as input for the optimization stage. The scheduling problem was formulated as a MILP, in which each binary variable indicates the selection of a pass. The objective function seeks to maximize the expected performance, defined as the duration of each contact multiplied by the predicted success probability and the estimated transmission rate, additionally penalizing a cost associated with the configuration time (setup) before and after the link. The restrictions imposed include the prohibition of undermining communication intervals, considering margins of ± 120 seconds to represent preparation times, as well as a capacity limit in the total scheduled time. The resolution of the model was carried out using the Coin-or branch and cut (CBC) solver through the PuLP library.

Finally, to evaluate the effectiveness of predictive policy, three reference strategies were defined. The first is a greedy

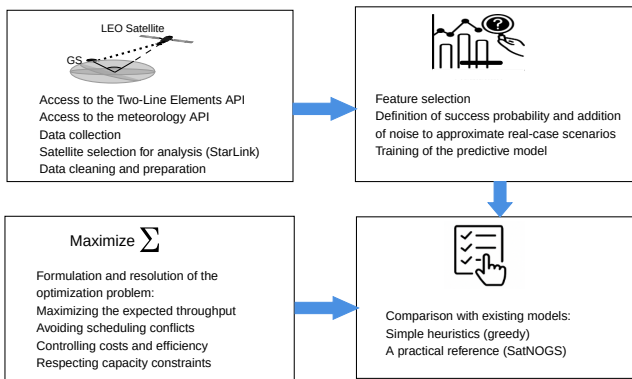


Fig. 2. Predictive Scheduling and Optimization for LEO Satellite Communications

were obtained using the Open-Meteo API, aligned with the duration of each pass over the GS [34]. From this information,

heuristic in chronological order, which selects the available passes based on their starting point, avoiding overlaps. The second is a greedy approach based on expected value, which prioritizes contacts with the highest individual expected throughput. The third one, inspired by the practice used by open networks like SatNOGS, selects all passes whose maximum elevation exceeds 20° without considering probabilities or configuration restrictions. The comparison between these approaches enabled us to quantify the improvement achieved by the proposed predictive strategy under realistic conditions of meteorological uncertainty.

IV. PREDICTIVE MODELING AND OPTIMIZATION FORMULATION

As mentioned in the previous section, a pass was considered successful when the maximum elevation angle exceeded 20° and the average precipitation was less than 0.5 mm/h, characterizing the problem as a binary classification. Therefore, a supervised learning approach was adopted to estimate the probability of successful transmission in each satellite pass. A Random Forest Classifier with 200 trees was trained using orbital, meteorological, and temporal features extracted from historical data. To ensure probabilistic reliability, the model was calibrated with CalibratedClassifierCV using a sigmoid method and 5-fold stratified cross-validation. Building upon the predictive model, the scheduling task was formulated as a MILP. For each pass i , we define a binary decision variable $x_i \in \{0, 1\}$, which indicates whether the pass is scheduled. The objective function maximizes the expected throughput while penalizing excessive scheduling through a setup cost:

$$\max \sum_i x_i \cdot p_i \cdot d_i \cdot R - C_{\text{setup}} \cdot \sum_i x_i, \quad (6)$$

where p_i is the predicted probability of success, d_i is the duration of pass i , R is the assumed transmission bitrate, and C_{setup} is the penalty cost associated with setup time.

The optimization is subject to three main constraints. First, non-overlap, which prohibits simultaneous scheduling of passes whose time intervals overlap when expanded by ± 300 seconds to account for setup requirements [38]. Second, a capacity constraint, limiting the total scheduled duration to

$$\sum_i d_i \cdot x_i \leq 7200 \text{ seconds}. \quad (7)$$

Thus, restricting the allocation to one hour of contacts. Third, an optional cap may be imposed to restrict the maximum number of passes, depending on operational considerations. The resulting MILP was solved using the CBC solver through the PuLP optimization library, enabling efficient computation of optimal predictive schedules under realistic operational restrictions.

V. RESULTS AND DISCUSSION

Our proposal, with a predictive approach and planning using the MILP method, is evaluated against three benchmarks. The first, Greedy by Start, consists of a sequential selection of non-overlapping passes in chronological order, always choosing

the earliest available opportunity. The second, Greedy by Expected Value, prioritizes passes according to their individual expected throughput, defined by the product of duration, bitrate, and predicted probability of success, resolving conflicts sequentially. The third, SatNOGS-like, follows a heuristic inspired by community-driven networks such as SatNOGS, in which all passes with maximum elevation greater than or equal to 20° are scheduled without considering probabilities, meteorological conditions, or setup costs.

Table I indicates the data and parameters used in this article to evaluate the proposed model.

TABLE I
EVALUATION CONFIGURATION USED TO TRAIN AND EVALUATE THE SCHEDULING MODEL.

Item	Value
Latitude	-23.21°
Longitude	-45.88°
Altitude	650 m
Historical data window	70 days
Prediction horizon	10 days
Transmission bitrate	1000 kbps
Setup time	120 s
simulation date	30/10/2025

Table II shows the performance metrics of the classifier on the test set. The model shows balanced performance across the two classes, achieving an overall accuracy of 0.96. Class 0 achieves a greater precision of 0.95, indicating a lower incidence of false positives, while class 1 shows a greater recall of 0.95, reflecting greater sensitivity in identifying this category. The macro and weighted averages remain consistent at 0.96, suggesting no relevant bias between classes and demonstrating that the model generalizes well over the test set.

TABLE II
PERFORMANCE METRICS OF THE CLASSIFIER ON THE TEST SET.

Class	Precision	Recall	F1-score	Support
0	0.95	0.97	0.96	38
1	0.98	0.95	0.96	42
Accuracy	—	—	0.96	80
Macro avg	0.96	0.96	0.96	80
Weighted avg	0.96	0.96	0.96	80

Fig. 3 shows the distribution of success probabilities estimated by the classifier for future passes. Note the predominant concentration in the range of 0.1 to 0.9, indicating that the model is rarely completely accurate. This dispersion shows that the model captures real uncertainties in the scenario. The result also indicates the need for further calibration, as intermediate probabilities make it difficult to make optimal decisions in programming policies.

On the other hand, Fig. 4 ship between predicted probabilities and actual labels. It can be seen that many events labeled as successful are not concentrated exclusively in high-probability regions, but are also distributed in intermediate ranges. Similarly, some events with a probability of failure appear in regions with a relatively high $P(\text{success})$. This pattern suggests a model with moderate discriminatory power:

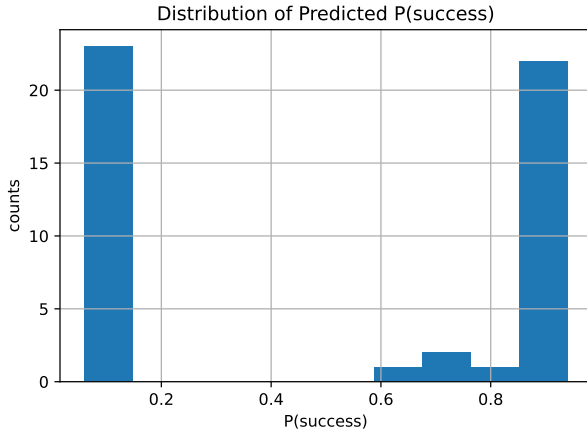


Fig. 3. Distribution of Predicted P(success)

there is a correlation between the predicted values and the labels, but significant overlaps persist.

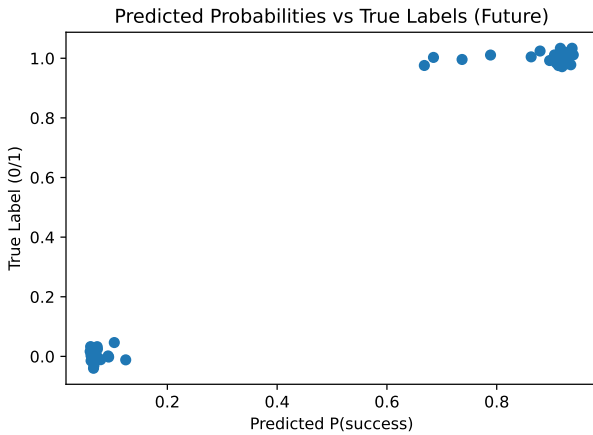


Fig. 4. Predicted Probabilities vs True Labels (Future)

Finally, Fig. 5 provides a temporal comparison of actual and predicted values across different passes. It should be noted that the model tends to oscillate around intermediate values between 0.1 and 0.3, with fewer occurrences of extreme predictions close to 0 or 1. In some cases, when the actual label is 1, the probability assigned by the model is also relatively high, demonstrating alignment. However, there are several cases where the predictions do not adequately match the actual behavior, highlighting the limitations of the model. This result reinforces the need for additional probabilistic calibration techniques or the use of more expressive models to improve reliability.

The predictive MILP policy consistently achieved higher realized throughput than the greedy baselines. Calibration was critical: uncalibrated models tended to overestimate throughput. The setup penalty effectively prevented inefficient allocation to short, low-value passes. The realized and expected throughput for each policy are summarized in Table III.

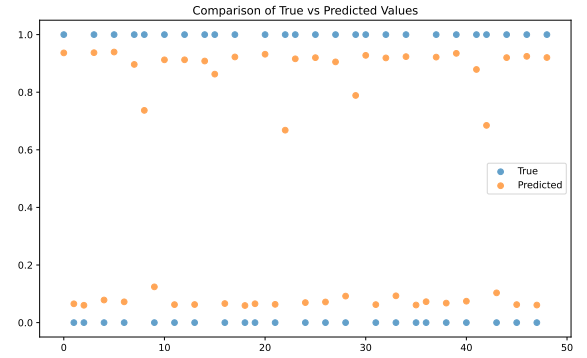


Fig. 5. Comparison of True vs Predicted Values

TABLE III
SCHEDULING RESULTS: EXPECTED AND REALIZED THROUGHPUT

Method	Expected (kB)	Realized (kB)
Predictive MILP	18,450	16,900
Greedy Start	16,200	14,500
Greedy Value	17,800	15,400
SatNOGS	20,100	13,200

VI. CONCLUSIONS

This study presents a method that integrates machine learning and mixed-integer linear programming to plan LEO satellite steps under uncertain weather conditions. This approach outperforms heuristic methods by using calibrated probability estimates to optimize planning, resulting in more robust and efficient ground station operations. Although the proposed approach shows significant improvements, several limitations must be considered. First, the method is based on the accuracy of weather forecasts, which directly affects predictive performance. Second, the scalability of MILP may be limited: the solver works well for dozens of passes, but may be computationally demanding for large constellations. Finally, the labeling of the pass success was based on proxy rules; actual RF link measurements may differ. In the future, this framework could be extended to multi-station scheduling, real-time online learning, and multi-objective optimization, considering both throughput and energy efficiency, service fairness, and payload priorities.

REFERENCES

- [1] A. Yuan, Z. Yang, and Z. Sun, "Evolution of satellite communication systems toward 5g/6g for 2030 and beyond," *Engineering*, 2025.
- [2] M. Höyhty, S. Boumard, A. Yastrebova, P. Järvensivu, M. Kiviranta, and A. Anttonen, "Sustainable satellite communications in the 6g era: A european view for multilayer systems and space safety," *IEEE Access*, vol. 10, pp. 99 973–100 005, 2022.
- [3] I. Ud Din, M. Guizani, S. Hassan, B.-S. Kim, M. Khurram Khan, M. Atiquzzaman, and S. H. Ahmed, "The internet of things: A review of enabled technologies and future challenges," *IEEE Access*, vol. 7, pp. 7606–7640, 2019.
- [4] J. Kua, S. W. Loke, C. Arora, N. Fernando, and C. Ranaweera, "Internet of things in space: A review of opportunities and challenges from satellite-aided computing to digitally-enhanced space living," *Sensors*, vol. 21, no. 23, p. 8117, 2021.

- [5] F. Vatalaro, G. Corazza, C. Caini, and C. Ferrarelli, "Analysis of leo, meo, and geo global mobile satellite systems in the presence of interference and fading," *IEEE Journal on Selected Areas in Communications*, vol. 13, no. 2, pp. 291–300, 1995.
- [6] F. S. Prol, R. M. Ferre, Z. Saleem, P. Välsuö, C. Pinell, E. S. Lohan, M. Elsanhoury, M. Elmusrati, S. Islam, K. Çelikbilek, K. Selvan, J. Yliaho, K. Rutledge, A. Ojala, L. Ferranti, J. Praks, M. Z. H. Bhuiyan, S. Kaasalainen, and H. Kuusniemi, "Position, navigation, and timing (pnt) through low earth orbit (leo) satellites: A survey on current status, challenges, and opportunities," *IEEE Access*, vol. 10, pp. 83 971–84 002, 2022.
- [7] H. Xie, Y. Zhan, G. Zeng, and X. Pan, "Leo mega-constellations for 6g global coverage: Challenges and opportunities," *IEEE Access*, vol. 9, pp. 164 223–164 244, 2021.
- [8] S.-A. Boukabara, J. Eyre, R. A. Anthes, K. Holmlund, K. M. St. Germain, and R. N. Hoffman, "The earth-observing satellite constellation: A review from a meteorological perspective of a complex, interconnected global system with extensive applications," *IEEE Geoscience and Remote Sensing Magazine*, vol. 9, no. 3, pp. 26–42, 2021.
- [9] X. Yang, X. Gao, H. Liu, X. Wu, and R. Han, "Availability analysis of remote sensing satellite systems using 25.5-27 ghz band for data transmission applications," in *2019 International Conference on Wireless Communication, Network and Multimedia Engineering (WCNME 2019)*. Atlantis Press, 2019, pp. 18–23.
- [10] T. Yanagisawa, H. Kurosaki, H. Oda, and M. Tagawa, "Ground-based optical observation system for leo objects," *Advances in Space Research*, vol. 56, no. 3, pp. 414–420, 2015.
- [11] T. E. Humphreys, P. A. Iannucci, Z. M. Komodromos, and A. M. Graff, "Signal structure of the starlink ku-band downlink," *IEEE Transactions on Aerospace and Electronic Systems*, vol. 59, no. 5, pp. 6016–6030, 2023.
- [12] A. M. Al-Saegh, A. Sali, J. Mandeep, and A. Ismail, "Extracted atmospheric impairments on earth-sky signal quality in tropical regions at ku-band," *Journal of Atmospheric and Solar-Terrestrial Physics*, vol. 104, pp. 96–105, 2013.
- [13] A. Maitra, A. De, and A. Adhikari, "Rain and rain-induced degradations of satellite links over a tropical location," *IEEE Transactions on Antennas and Propagation*, vol. 67, no. 8, pp. 5507–5518, 2019.
- [14] G. Zeng, Y. Zhan, H. Xie, and C. Jiang, "Resource allocation for networked telemetry system of mega leo satellite constellations," *IEEE Transactions on Communications*, vol. 70, no. 12, pp. 8215–8228, 2022.
- [15] F. Chai, Q. Zhang, H. Yao, X. Xin, R. Gao, and M. Guizani, "Joint multi-task offloading and resource allocation for mobile edge computing systems in satellite iot," *IEEE Transactions on Vehicular Technology*, vol. 72, no. 6, pp. 7783–7795, 2023.
- [16] S. Wang, X. Tang, J. Li, X. Huang, J. Liu, and J. Liu, "Navigation resource allocation algorithm for leo constellations based on dynamic programming," *Remote Sensing*, vol. 16, no. 12, 2024. [Online]. Available: <https://www.mdpi.com/2072-4292/16/12/2231>
- [17] M. Lv, X. Peng, W. Xie, and N. Guan, "Task allocation for real-time earth observation service with leo satellites," in *2022 IEEE Real-Time Systems Symposium (RTSS)*, 2022, pp. 14–26.
- [18] L. Zhong, Y. Li, M.-F. Ge, M. Feng, and S. Mao, "Joint task offloading and resource allocation for leo satellite-based mobile edge computing systems with heterogeneous task demands," *IEEE Transactions on Vehicular Technology*, vol. 74, no. 7, pp. 11 337–11 352, 2025.
- [19] W. Usaha and J. A. Barria, "Reinforcement learning for resource allocation in leo satellite networks," *IEEE Transactions on Systems, Man, and Cybernetics, Part B (Cybernetics)*, vol. 37, no. 3, pp. 515–527, 2007.
- [20] V. F. Adanvo, S. Mafra, S. Montejó-Sánchez, F. A. Tondo, and R. D. Souza, "Improving efficiency and reliability in information offloading for leo satellite networks by inter-satellite communication techniques," *IEEE Transactions on Aerospace and Electronic Systems*, 2024.
- [21] Y. Zhan, P. Wan, C. Jiang, X. Pan, X. Chen, and S. Guo, "Challenges and solutions for the satellite tracking, telemetry, and command system," *IEEE Wireless Communications*, vol. 27, no. 6, pp. 12–18, 2021.
- [22] Z. M. Kassas, S. Hayek, and J.-H. Ahmad, "Leo satellite orbit prediction via closed-loop machine learning with application to opportunistic navigation," *IEEE Aerospace and Electronic Systems Magazine*, vol. 40, no. 1, pp. 34–49, 2025.
- [23] I. Ahmad, S. Shahabuddin, H. Malik, E. Harjula, T. Leppänen, L. Lovén, A. Anttonen, A. H. Sodhro, M. Mahtab Alam, M. Juntti, A. Ylä-Jääski, T. Sauter, A. Gurtov, M. Ylianttila, and J. Riekk, "Machine learning meets communication networks: Current trends and future challenges," *IEEE Access*, vol. 8, pp. 223 418–223 460, 2020.
- [24] S. Salcedo-Sanz, J. Pérez-Aracil, G. Ascenso, J. Del Ser, D. Casillas-Pérez, C. Kadow, D. Fister, D. Barriopedro, R. García-Herrera, M. Giuliani *et al.*, "Analysis, characterization, prediction, and attribution of extreme atmospheric events with machine learning and deep learning techniques: a review," *Theoretical and Applied Climatology*, vol. 155, no. 1, pp. 1–44, 2024.
- [25] A. Akbari Asanjan, T. Yang, K. Hsu, S. Sorooshian, J. Lin, and Q. Peng, "Short-term precipitation forecast based on the persiann system and lstm recurrent neural networks," *Journal of Geophysical Research: Atmospheres*, vol. 123, no. 22, pp. 12–543, 2018.
- [26] G. Ayzel, T. Scheffer, and M. Heistermann, "Rainnet v1. 0: a convolutional neural network for radar-based precipitation nowcasting," *Geoscientific Model Development*, vol. 13, no. 6, pp. 2631–2644, 2020.
- [27] Y. Zhang, Y. Wu, A. Liu, X. Xia, T. Pan, and X. Liu, "Deep learning-based channel prediction for leo satellite massive mimo communication system," *IEEE Wireless Communications Letters*, vol. 10, no. 8, pp. 1835–1839, 2021.
- [28] K. Çelikbilek, Z. Saleem, R. Morales Ferre, J. Praks, and E. S. Lohan, "Survey on optimization methods for leo-satellite-based networks with applications in future autonomous transportation," *Sensors*, vol. 22, no. 4, 2022. [Online]. Available: <https://www.mdpi.com/1424-8220/22/4/1421>
- [29] F. Valente, F. G. Lavacca, M. Polverini, T. Fiori, and V. Eramo, "Optimization of ground station energy saving in leo satellite constellations for earth observation applications," *Future Internet*, vol. 17, no. 6, 2025. [Online]. Available: <https://www.mdpi.com/1999-5903/17/6/229>
- [30] G. Zhao, Z. Kang, Y. Huang, and S. Wu, "A routing optimization method for leo satellite networks with stochastic link failure," *Aerospace*, vol. 9, no. 6, p. 322, 2022.
- [31] J. Qi, J. Guo, M. Wang, and C. Wu, "A cooperative autonomous scheduling approach for multiple earth observation satellites with intensive missions," *IEEE Access*, vol. 9, pp. 61 646–61 661, 2021.
- [32] D.-H. Cho, J.-H. Kim, H.-L. Choi, and J. Ahn, "Optimization-based scheduling method for agile earth-observing satellite constellation," *Journal of Aerospace Information Systems*, vol. 15, no. 11, pp. 611–626, 2018.
- [33] Skyfielders, "python-skyfield: Elegant astronomy for python," 2025, repositorio GitHub, licença MIT. [Online]. Available: <https://github.com/skyfielders/python-skyfield>
- [34] P. Zippenfenig and colaboradores, "Open-Meteo.com Weather API," 2024, aPI meteorológica aberta — dados sob licença CC BY 4.0. [Online]. Available: <https://open-meteo.com/>
- [35] K. Tsutsui, S. Rokugawa, H. Nakagawa, S. Miyazaki, C.-T. Cheng, T. Shiraishi, and S.-D. Yang, "Detection and volume estimation of large-scale landslides based on elevation-change analysis using dems extracted from high-resolution satellite stereo imagery," *IEEE Transactions on Geoscience and Remote Sensing*, vol. 45, no. 6, pp. 1681–1696, 2007.
- [36] U. (UIT-R), "Características de los sistemas de satélite que se han de tener en cuenta en los análisis de compartición de frecuencias entre sistemas de satélites geoestacionarios (osg) y no osg del servicio fijo por satélite," Sector de Radiocomunicaciones de la UIT (UIT-R), Ginebra, Suiza, Recomendación S.1328-3, febrero 2001. [Online]. Available: https://www.itu.int/dms_pubrec/itu-r/rec/s/R-REC-S.1328-3-200102-S!!PDF-S.pdf
- [37] International Telecommunication Union (ITU), "Characteristics of precipitation for propagation modelling," ITU Radiocommunication Sector (ITU-R), Geneva, Switzerland, Recommendation P.837-8, September 2025, available at: <https://www.itu.int/rec/R-REC-P.837-8-202509-I/en>. [Online]. Available: <https://www.itu.int/rec/R-REC-P.837-8-202509-I/en>
- [38] M. Schmidt, M. Rybysc, and K. Schilling, "A scheduling system for small ground station networks," in *SpaceOps 2008 conference*, 2008, p. 3578.